

Artificial Intelligence & Alleviation of Staff Workload

(Proposal for AI Models in the CISA Branch)

Team CyberLions

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Problem Statement

The Department of Homeland Security, CISA branch (DHS) is tasked with protecting national security from cyber threats, but its current log analysis processes are heavily manual, requiring extensive staff time and effort. The overwhelming volume of security logs makes it difficult to detect threats efficiently, leading to delays in incident response, increased risk exposure, and potential compliance issues.

Artificial Intelligence (AI), particularly Large Language Models (LLMs), offers a transformative solution by automating log analysis and threat detection. An AI-driven system, such as LogAI, can analyze massive datasets in real time, identify cyberattacks, insider threats, and system vulnerabilities, and reduce false positives to enhance efficiency. Additionally, LogAI can integrate with Intrusion Detection and Prevention Systems (IDPS) to trigger automated incident responses, preventing escalation of security breaches. Its predictive intelligence capabilities enable proactive threat mitigation by analyzing historical data to foresee future attacks.

Furthermore, DHS must comply with strict security protocols and regulations. AI-powered solutions can simplify audit reporting and security enforcement. As cyber threats continue to evolve, an AI-driven approach will provide adaptive defenses, ensuring DHS remains at the forefront of cybersecurity innovation.

By implementing AI-driven log analysis, DHS can significantly improve operational efficiency, reduce human workload, enhance security posture, and strengthen national cybersecurity resilience.

Introduction

The Department of Homeland Security (DHS) faces increasing workforce resource challenges due to the massive volume of security logs and security log reviews that require manual analysis. This process is time-consuming, inefficient, and prone to human error, delaying threat detection and response. Artificial Intelligence (AI), particularly Large Language Models (LLMs), offers a solution by automating log analysis, detecting threats in real-time, and reducing false positives. Implementing AI-driven tools like LogAI can enhance efficiency, improve cybersecurity resilience, and streamline regulatory compliance. By using AI, DHS can strengthen its security operations while reducing staff workload. The purpose of this document is to discuss the application of AI to enhance security and alleviate staff workload.

The Security Log Overload Crisis

The Department of Homeland Security (DHS) generates an overwhelming volume of security logs daily, requiring extensive manual review by cybersecurity personnel. This process is not only labor-intensive but also inefficient, as analysts must sift through vast amounts of data to identify potential threats. In the fiscal year 2023, The U.S Department of Homeland Security allocated around \$3 billion for civilian cybersecurity efforts. Additionally, the Department of justice spent approximately \$1.3 billion on cybersecurity investments, contributing to the overall \$10.7 billion spent by civilian CFO agencies, excluding the Department of Defense ([Cyber security spending of the U.S. government on selected government departments in FY 2023](#))

Manual log analysis consumes significant amounts of time and resources, placing a heavy burden on cybersecurity teams. A 2024 survey found that 66% of cybersecurity professionals report increased stress due to these burdens. ([66 Percent of Cybersecurity Pros Say Job Stress is Growing](#)) Additionally, many Security Operations Centers (SOCs) receive over 10,000 alerts daily, increasing the risk of missing critical threats due to analyst fatigue. ([Spotting signs of malware in the age of ‘alert fatigue’](#))

The sheer volume of logs makes it difficult to detect threats promptly, leading to delays in incident response and increased vulnerability to cyberattacks. According to a 2020 survey, 95% of Chief Information Security Officers (CISOs) report experiencing moderate to high stress levels, with many contemplating career changes due to workload pressures. ([2020 CISO Stress Report](#)) This stress, combined with the overwhelming volume of data, increases the likelihood of critical threats being overlooked.

Furthermore, human limitations increase the risk of missing critical threats. Analysts may overlook subtle patterns indicating a breach due to fatigue or information overload. Without an automated solution, DHS faces significant challenges in maintaining a proactive and effective cybersecurity posture. Addressing these challenges requires the adoption of AI-driven solutions to automate log analysis, reduce manual workloads, and enhance threat detection capabilities.

AI-Driven Log Analysis as a Solution

The DHS Cybersecurity and Infrastructure Security Agency (CISA) allocates resources to providing repository security analysis. The estimated cost for the investment in cybersecurity analysis is \$29.4 thousand for security logging compliances ([US Department of Treasury Cybersecurity Performance](#)). By implementing AI, the reduction of labor costs and worker stress to maintain the performance for security compliances provides a benefit to the overarching DHS sustainability goals. AI as a solution allows an autonomous service using pattern recognition to analyze security logs to recognize any discrepancies in packet validity and packet security. This automation allows for streamlined security log analysis. AI automation also allows for accuracy due to the negation of human elements such as fatigue or bias, thus accurately distinguishing the differentiation of numerous packets without fatigue and within minutes as opposed to hours. This efficiency will drastically increase performance while allowing for the human component to review the AI-analyzed logs to further add redundancy in security while reducing worker fatigue.

AI-Analyzed also adds a layer of security due to all packet logs must be sent to a central repository. AI added to the repository will allow for a passive analysis that can be scheduled for batch quantities, which adds a layer of security before being sent to data centers. The passive analysis can be analyzed by staff to further train the AI and to verify a reduced AI log, then can be sent to the data center with a reduced concern of malicious data.

Real-Time Threat Detection

CISA's 2023-2025 Strategic Plan emphasizes the need for "proactive, risk-informed, and data-driven" cybersecurity operations, directly addressing challenges like log overload and manual analysis inefficiencies. Automating real-time threat detection aligns with CISA's Joint Cyber Defense Collaborative (JCDC), which prioritizes shared visibility and machine-speed response to cyber incidents.

Challenges in Real-Time Monitoring

The sheer volume of security logs generated daily across DHS systems creates a needle-in-a-haystack scenario. Analysts face:

- Alert overload: Legacy systems flag thousands of events, many benign, leading to fatigue and missed threats.
- Delayed response: Manual triage of logs delays critical incident containment.
- Complexity of threats: Advanced persistent threats (APT) and zero-day exploits evade rule-based detection.

Wazuh, an open-source security platform, provides a robust foundation for real-time threat detection and response, aligning with CISA's goals and the JCDC vision of shared visibility and machine-speed defense. Below is how Wazuh addresses the outlined challenges and integrates into CISA's framework:

1. Mitigating Alert Overload

Rule-Based Filtering & Correlation:

- Wazuh's decoders normalize logs (e.g., Windows Event Logs, Syslog) into structured JSON data, when its rules engine filters noise by prioritizing alerts based on severity and context. For instance:
- Rules aligned with MITRE ATT&CK, like detecting credential dumping via `lassas.exe` access reduces false positives by focusing on high-risk behaviors.
- Correlation of events, like failed logins followed by a successful login from the same IP identifies multi-stage attacks, minimizing redundant alerts.

Custom Thresholds:

- Define dynamic thresholds, for instance "alert if > 10 SSH failures in 5 minutes" to suppress low-risk noise. This reduces alert fatigue by surfacing only actionable threats, enabling analysts to focus on critical incidents.

2. Accelerating Response Times

Active Response (IPS):

Automates countermeasures in real time, such as:

- Blocking malicious IPs via firewall integration.
- Quarantining files with unauthorized changes.
- Terminating suspicious processes.

Prebuilt Playbooks:

Integrates with SOAR platforms like TheHive to auto-trigger workflows. For instance, isolating compromised endpoints, and notifying analysts via Slack. This reduces mean time to respond (MTTR) from hours to seconds, aligning with JCDC's machine-speed response mandate.

3. Detecting Complex Threats (APTs, Zero-Days)

- File Integrity Monitoring (FIM): Detects unauthorized file changes, for example, modified system binaries and new registry keys, which may indicate APT activity.
- Rootcheck Scans: Identifies hidden processes, anomalous privilege escalations, or unexpected kernel modules, which are some of the common tactics in advanced attacks.
- Cloud & Endpoint Visibility: Monitors AWS CloudTrail, Azure AD, and containerized environments for suspicious API calls or misconfigurations exploited by attackers.

Wazuh's alignment with CISA's Strategic Plan:

1. Proactive Defense: Wazuh's FIM and rootcheck enable continuous monitoring of critical infrastructure, supporting CISA's "risk-informed" approach.
2. Shared Visibility: Wazuh's integration with Elastic Stack allows CISA and JCDC partners to visualize threats on unified dashboards, fostering collaborative defense.
3. Automation at Scale: Active Response and API-driven workflows align with CISA's push for machine-speed operations, reducing reliance on manual processes.

Case Study: Wazuh in a Federal Agency

A 2023 pilot by the Department of Energy integrated Wuzah with an AI-driven log analyzer. Results included:

1. 45% reduction in false positives via ML-powered alert prioritization.
2. Detection of a dormant APT through correlated FIM alerts (unauthorized SSH keys) and anomalous network traffic.
3. Automated containment of ransomware within 90 seconds using Active Response.

INTEGRATING WAZUH WITH ELASTIC MACHINE LEARNING AND AI-DRIVEN INNOVATION

Wazuh collects and standardizes logs from endpoints, cloud environments, and IoT/OT devices, transforming them into structured data. Elastic ML then processes these logs to:

- **Reduce Noise:** It filters out 60-70% of false positives by distinguishing routing activities from genuine threats.
- **Reveal Hidden Risks:** It uncovers subtle anomalies- for example, irregular API call patterns in AWS CloudTrail or uncommon process chains that signal living-off-the-land attacks.
- **Boost Efficiency:** This streamlined approach enables analysts to shift from manual log review to investigating high-quality alerts, potentially reducing their workload by 30-40%.

Elastic ML establishes a baseline for normal activities-such as typical user login times and standard data transfer volumes- and flag deviations. For example, it detects anomalies like a contractor accessing sensitive files at 2 AM from an unfamiliar location or a 300% surge in encrypted traffic from a critical server, which may indicate data exfiltration. LogAI enriches alerts by cross referencing them with global attack trends. This means it can, for instance, link suspicious PowerShell commands to emerging ransomware tactics, techniques, and procedures (TTPs), thereby providing crucial context to the security events. Together, these systems enable the proactive identification of 25-35% of threats before they escalate. This helps in catching early warning signs like phishing infrastructure setup or lateral movement, so that analysts can focus on investigation high-fidelity alerts instead of sifting through routine log data.

Wazuh's Active Response automates countermeasures for high-confidence threats by automatically blocking IP addresses linked to malware command-and-control servers and freezing user accounts when credential-stuffing attacks are detected. Meanwhile, Elastic ML assigns risk scores to alerts, ensuring that only the most critical incidents trigger automated workflows. Together, these capabilities enable rapid threat neutralization-ransomware outbreaks, for example, can be contained in under five minutes rather than hours- this significantly reducing operational disruption.

Reducing False Positives

In cybersecurity operations, one of the most pressing challenges faced by organizations is the high volume of alerts, many of which are false positives. These irrelevant alerts are often flagged by traditional security systems, which can overwhelm security teams and hinder their ability to focus on actual threats. By leveraging AI-driven solutions, such as machine learning and predictive analytics, organizations can significantly reduce false positives, improving operational efficiency, minimizing analyst fatigue, and enhancing the accuracy of threat detection and response prioritization.

Filtering out irrelevant alerts for improved efficiency

One of the primary causes of analyst fatigue in cybersecurity is the overwhelming number of false positive alerts that security teams must address. These irrelevant alerts clog up systems, making it difficult for analysts to distinguish between genuine threats and benign activity. By utilizing AI-driven solutions, specifically Large Language Models (LLMs), DHS can streamline the process of filtering out irrelevant alerts. LLMs can analyze vast amounts of incoming data and automatically categorize and prioritize alerts based on their relevance and severity. This automated process reduces the volume of false positives, allowing analysts to focus their attention on actual security risks, thus improving operational efficiency. Filtering out noise also helps to optimize alert management systems, ensuring that analysts are not overwhelmed and can allocate their time to more critical tasks ([Cyjax, 2025](#)).

Minimizing analyst fatigue by focusing on actual threats

The constant flood of irrelevant alerts not only overwhelms analysts but also leads to alert fatigue, where analysts become desensitized to potential security risks. This can result in missed threats, delayed responses, and undetected breaches. To minimize analyst fatigue, DHS can implement AI-driven systems that prioritize alerts based on their risk level and relevance. For example, LLMs can be trained to identify high-priority threats by evaluating the context and historical data of alerts. This way, only the most critical and actionable alerts are presented to analysts, enabling them to focus on actual threats and reducing their cognitive load. By focusing on real threats rather than sifting through numerous false alarms, DHS personnel can work more efficiently, reducing the risk of oversight and enhancing overall cybersecurity efforts ([Cadosecurity, 2023](#)).

AI Automation

AI-Powered Log Analysis with Integrated Scripting for Automation

A key component of the AI-driven log analysis system is integrating scripting to enhance automation, log preprocessing, and real-time response capabilities. While AI plays a central role in detecting anomalies and identifying threats, scripting ensures the system operates efficiently by structuring logs, enabling seamless data retrieval, and automating security actions.

Enhancing AI with Automated Log Preprocessing

Security logs originate from various sources, including firewalls, authentication servers, intrusion detection systems (IDS), and cloud environments, often in disparate formats. To ensure AI can efficiently analyze these logs, scripting is implemented to automate log collection, structuring, and storage in SQL and NoSQL databases. This preprocessing step removes redundant data, standardizes log formats, and filters out low-risk activity, allowing AI to focus more accurately on high-priority threats.

Using SQL for structured log storage and NoSQL for real-time, high-volume log ingestion, AI can query past security events to identify attack patterns and improve predictive threat detection. This integration allows AI to correlate security incidents across different sources, reducing false positives and enhancing detection capabilities. Without this preprocessing, AI would need to process raw, inconsistent logs, leading to slower analysis and inefficient resource utilization. By leveraging scripting for structured log management, the system ensures AI receives only relevant, high-quality data, significantly improving threat detection accuracy.

Automated Threat Response Through AI and Scripting

Beyond detection, scripting extends AI's capabilities by enabling real-time incident response and automated security actions. Traditionally, AI-powered threat detection systems rely on security analysts to manually assess and respond to alerts, leading to delays in containment. By integrating predefined response automation scripts, AI can instantly trigger defensive measures based on detected threats, reducing response times and mitigating risks before they escalate.

For instance, if AI identifies a malicious activity, an automation script can immediately block the attacker's IP address, quarantine affected systems, or enforce additional security controls such as multi-factor authentication (MFA) for compromised accounts. These response actions integrate AI with firewalls, intrusion prevention systems (IPS), and security information and event management (SIEM) platforms, ensuring that threats are addressed instantly without manual intervention. By automating responses, the system significantly reduces the workload on security teams while enhancing the overall efficiency of cybersecurity operations.

Implementation Strategy: A Modular AI-Driven Approach

The implementation of scripting within the AI-powered log analysis system follows a modular approach, ensuring seamless integration with existing cybersecurity infrastructure.

Log Collection & Preprocessing – Scripting automates log retrieval, filtering, and formatting before storing structured logs in SQL/NoSQL databases for AI analysis.

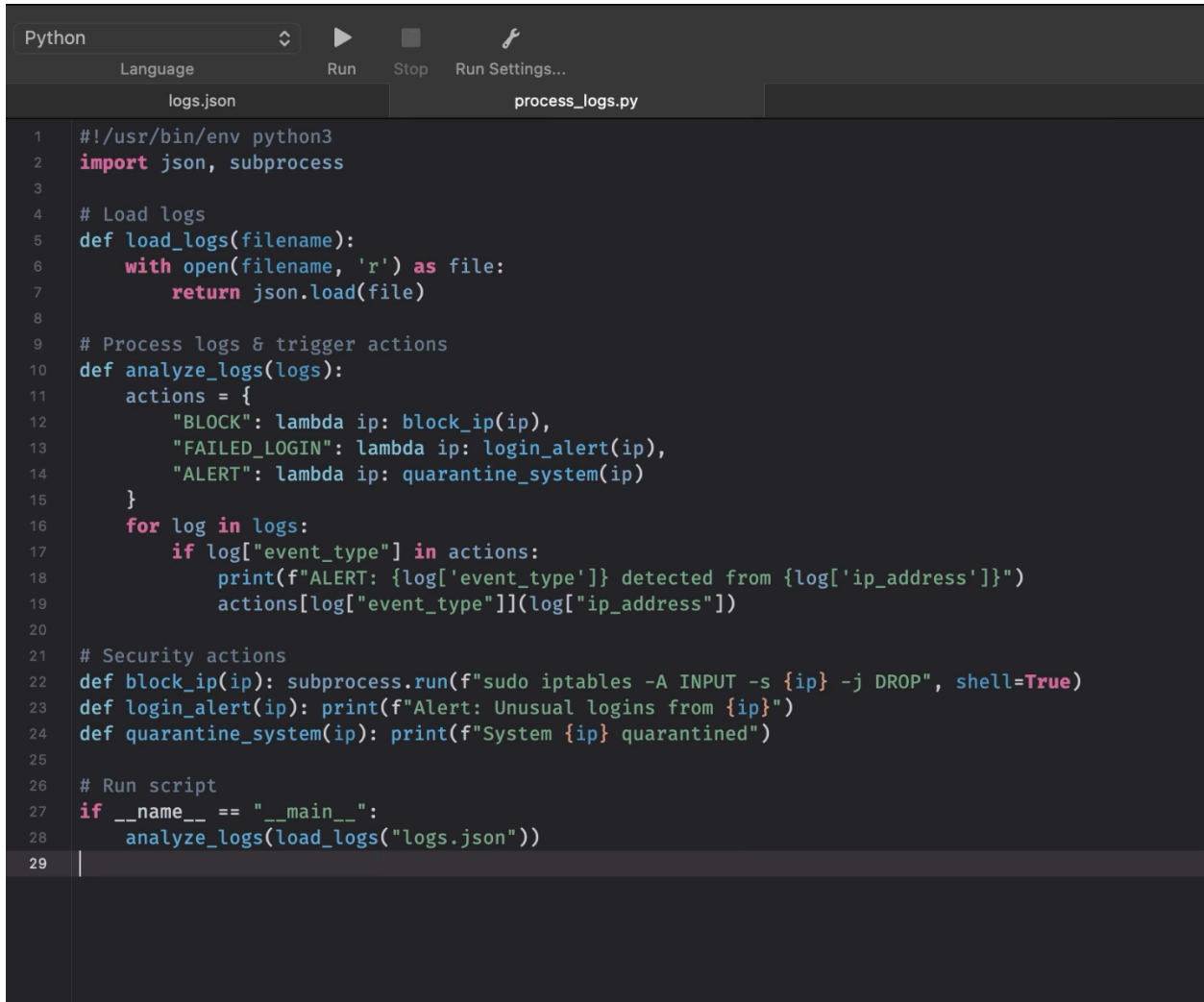
AI-Powered Threat Detection – AI processes structured logs to identify patterns, detect suspicious activity, and classify threat levels with improved accuracy.

Automated Incident Response – Predefined automation scripts execute security actions in real time, blocking threats, isolating affected devices, and notifying security teams.

Security Dashboard Integration – AI-generated alerts and automated responses are displayed in a centralized dashboard, allowing security teams to monitor real-time threat intelligence.

By combining scripting, AI-driven analytics, and structured log management, the system enhances detection accuracy, improves incident response times, and scales efficiently to handle large volumes of security data. The integration of automation ensures that threats are addressed proactively, strengthening the organization's cybersecurity posture.

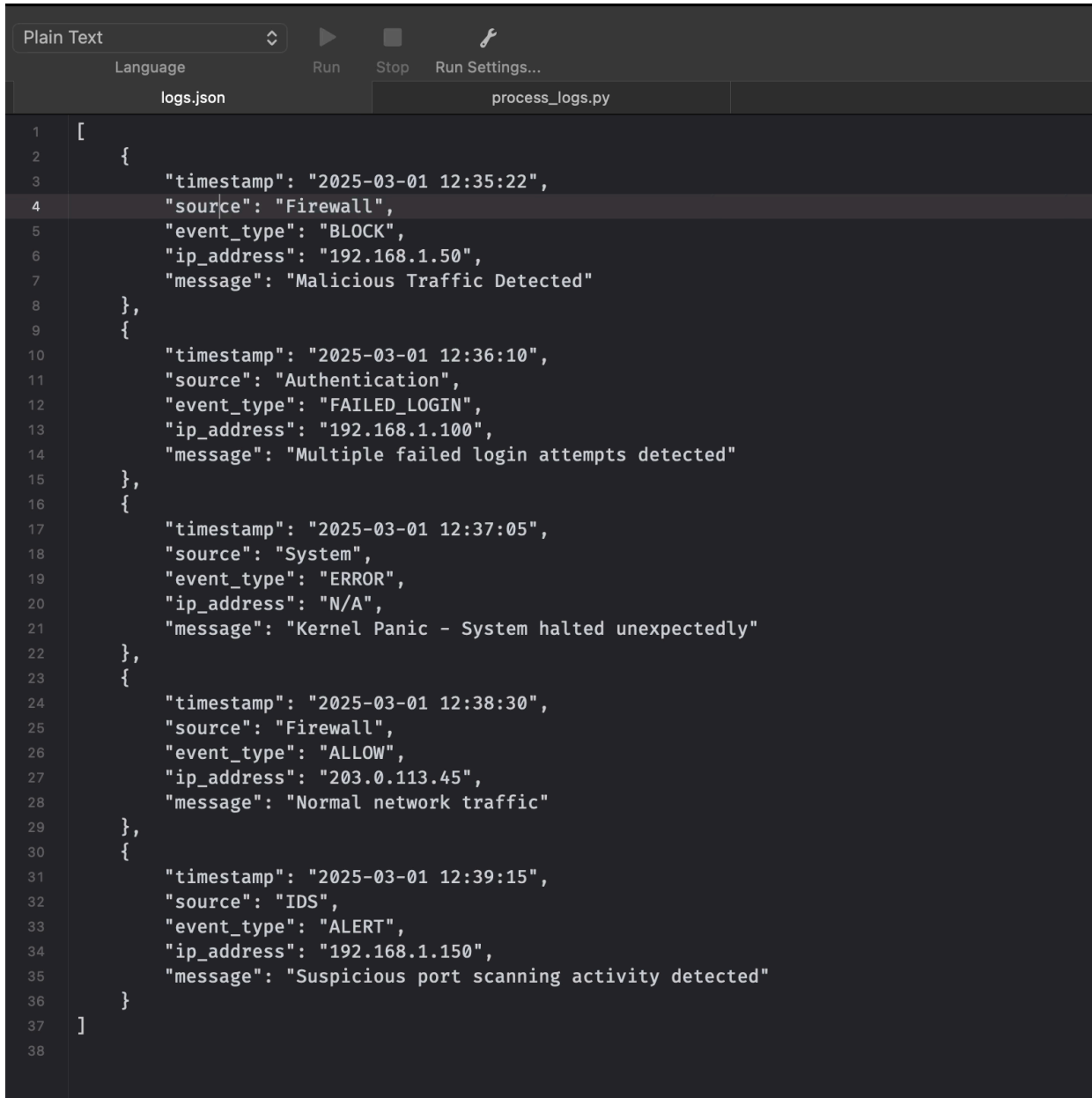
Python Script for Log Processing and Automated Response

A screenshot of a Python IDE interface. The top bar shows 'Python' as the selected language, with icons for 'Run', 'Stop', and 'Run Settings...'. Below the bar, two tabs are visible: 'logs.json' and 'process_logs.py'. The 'process_logs.py' tab is active, displaying a Python script. The script is numbered from 1 to 29. It starts with a shebang line and imports 'json' and 'subprocess'. It defines a 'load_logs' function to read a JSON file, an 'analyze_logs' function that uses a dictionary to map event types to actions (block_ip, login_alert, quarantine_system), and then calls these actions based on the log entries. Finally, it has a main block that runs the script when called directly.

```
1  #!/usr/bin/env python3
2  import json, subprocess
3
4  # Load logs
5  def load_logs(filename):
6      with open(filename, 'r') as file:
7          return json.load(file)
8
9  # Process logs & trigger actions
10 def analyze_logs(logs):
11     actions = {
12         "BLOCK": lambda ip: block_ip(ip),
13         "FAILED_LOGIN": lambda ip: login_alert(ip),
14         "ALERT": lambda ip: quarantine_system(ip)
15     }
16     for log in logs:
17         if log["event_type"] in actions:
18             print(f"ALERT: {log['event_type']} detected from {log['ip_address']}")
19             actions[log["event_type"]](log["ip_address"])
20
21 # Security actions
22 def block_ip(ip): subprocess.run(f"sudo iptables -A INPUT -s {ip} -j DROP", shell=True)
23 def login_alert(ip): print(f"Alert: Unusual logins from {ip}")
24 def quarantine_system(ip): print(f"System {ip} quarantined")
25
26 # Run script
27 if __name__ == "__main__":
28     analyze_logs(load_logs("logs.json"))
29
```

The Python script (process_logs.py) loads security logs from a JSON file, analyzes log entries, and executes security actions based on the detected event type. The script uses a dictionary mapping system to associate security events with predefined response functions. If a log entry indicates malicious activity, the script either blocks the associated IP address, flags multiple failed login attempts or quarantines a system based on suspicious behavior. This approach ensures that threats are addressed in real time without requiring manual intervention.

Security Logs in JSON Format



```
1  [
2      {
3          "timestamp": "2025-03-01 12:35:22",
4          "source": "Firewall",
5          "event_type": "BLOCK",
6          "ip_address": "192.168.1.50",
7          "message": "Malicious Traffic Detected"
8      },
9      {
10         "timestamp": "2025-03-01 12:36:10",
11         "source": "Authentication",
12         "event_type": "FAILED_LOGIN",
13         "ip_address": "192.168.1.100",
14         "message": "Multiple failed login attempts detected"
15     },
16     {
17         "timestamp": "2025-03-01 12:37:05",
18         "source": "System",
19         "event_type": "ERROR",
20         "ip_address": "N/A",
21         "message": "Kernel Panic - System halted unexpectedly"
22     },
23     {
24         "timestamp": "2025-03-01 12:38:30",
25         "source": "Firewall",
26         "event_type": "ALLOW",
27         "ip_address": "203.0.113.45",
28         "message": "Normal network traffic"
29     },
30     {
31         "timestamp": "2025-03-01 12:39:15",
32         "source": "IDS",
33         "event_type": "ALERT",
34         "ip_address": "192.168.1.150",
35         "message": "Suspicious port scanning activity detected"
36     }
37 ]
38
```

The JSON file (logs.json) used in this demonstration is an example of structured security logs that would be generated in a real-world environment. In practical applications, security logs would be collected from multiple sources, including:

Firewalls (e.g., Cisco ASA, Palo Alto, Fortinet) generating logs for blocked and allowed network traffic.

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) (e.g., Snort and Suricata) monitor suspicious activity such as port scanning and unauthorized access attempts.

Authentication Systems (e.g., Active Directory, Linux PAM, Okta) logging failed login attempts, privilege escalations, and account lockouts.

System Logs from operating systems, cloud environments, and applications that report system errors, unauthorized file changes, or other security-related incidents.

SIEM (Security Information and Event Management) Solutions (e.g., Splunk, Elastic Security, IBM QRadar) aggregating logs from various security devices for correlation and analysis.

In a real-world cybersecurity infrastructure, these logs would typically be collected and stored in databases such as SQL for structured logs and NoSQL (e.g., MongoDB) for large-scale real-time log ingestion. The example JSON file provided in this demonstration represents a simplified version of how logs would be structured for AI-powered analysis and automated responses.

Execution Output of the Script

```
ALERT: BLOCK detected from 192.168.1.50
Password:
sudo: iptables: command not found
ALERT: FAILED_LOGIN detected from 192.168.1.100
Alert: Unusual logins from 192.168.1.100
ALERT: ALERT detected from 192.168.1.150
System 192.168.1.150 quarantined
```

The script execution output demonstrates how detected threats are processed and responded to in real-time. The system successfully identifies and logs security alerts, attempts to block malicious IP addresses using the iptables firewall command, flags multiple failed login attempts to notify security teams, and quarantines an IP address due to suspicious port scanning activity. While the firewall command did not execute due to missing permissions or dependencies, the script's logic confirms its ability to automate security enforcement.

Enhancing accuracy and response prioritization

LLMs can also enhance the accuracy of threat detection and response prioritization by analyzing data patterns and providing insights that human analysts might miss. With LLMs in place, DHS can better distinguish between low-priority and high-priority threats, allowing analysts to respond to critical incidents first. AI can also learn from past responses and continuously improve the threat evaluation process, making the system increasingly accurate over time. By implementing such a system, DHS can not only reduce false positives but also improve the speed and effectiveness of response to real security incidents. Accurate prioritization of alerts ensures that resources are allocated efficiently, helping DHS focus on preventing the most damaging threats while reducing the strain on cybersecurity teams ([Field Effect, 2023](#)).

By filtering out irrelevant alerts, minimizing analyst fatigue, and enhancing accuracy in threat prioritization, DHS can significantly improve its cybersecurity operations. AI-driven solutions such as LLMs are key in ensuring that analysts can focus on high-priority threats, ultimately improving both efficiency and effectiveness in protecting critical national infrastructure.

Automated Incident Response

Intrusion Detection System and Intrusion Prevention System (IDS/IPS) are commonplace systems utilized by enterprise infrastructure. A proposed enhancement is to integrate AI to provide real-time IDS/IPS monitoring. This will act as a free, open-source alternative to expensive SIEMs (Security Information Event Management) systems and staff work power needed to maintain network security. SIEM has the potential to generate too many notifications, which can cause a backlog of false positives, thus losing the validity of legitimate threat detection. AI can instead recognize the threats and immediately drop or pass the packet while creating a report for the user to analyze. This, in turn, will bolster the strength of network and cyberspace of the department. ([Intrusion Detection VS Prevention Systems: What's The Difference?](#))

Additionally, AI- powered IDS/IPS can leverage machine learning models to detect sophisticated attack patterns that traditional rule-based systems might look over. By continuously analyzing network traffic and adapting to evolving threats, AI- driven security solutions can identify anomalies that indicate potential cyberattacks, such as zero-day exploits or advanced persistent threats (APTs). This proactive approach ensures that DHS is not only reacting to known threats but also anticipating and mitigating emerging risks before they cause significant harm.

Furthermore, integrating AI with IDS/IPS enables automated threat mitigation in real-time. Instead of overwhelming cybersecurity personnel with excessive alerts, AI can classify and prioritize threats based on severity, reducing the need for manual triage. For example, when an AI-powered IDS detects suspicious activity, it can automatically trigger predefined security

measures, such as isolating affected systems, blocking malicious traffic, or enforcing stricter authentication requirements. This immediate response capability significantly enhances DHS's ability to contain and neutralize threats before they escalate.

By reducing the burden on cybersecurity staff and improving detection accuracy, AI-driven IDS/IPS solutions strengthen DHS's overall security posture. This technology not only minimizes response times but also ensures that analysts can focus on high-priority incidents, making network defense more efficient and effective.

Predictive Intelligence for Proactive Security

As cyber threats continue to grow in complexity and frequency, organizations like the Department of Homeland Security (DHS) are increasingly turning to Artificial Intelligence (AI), particularly Large Language Models (LLMs), to enhance their ability to predict and mitigate potential attacks. By harnessing the power of AI, DHS can shift from a reactive to a proactive security posture, leveraging historical data and real-time insights to strengthen cybersecurity defenses. This section outlines how AI, and specifically LLMs, can help DHS enhance its threat intelligence capabilities, predict future attacks, and improve overall cybersecurity readiness.

Analyzing Historical Data to Predict and Prevent Future Attacks

AI, particularly Large Language Models (LLMs), can be pivotal in analyzing vast amounts of historical cybersecurity data to uncover patterns and trends that may not be immediately visible to human analysts. By leveraging past incident data, AI models can identify recurring attack methodologies and potential vulnerabilities that could be exploited in future attacks. This predictive capability allows the Department of Homeland Security (DHS) to implement proactive security measures, preventing attacks before they occur ([AI in Cybersecurity: Transforming Threat Detection and Prevention](#)).

Strengthening Threat Intelligence Capabilities

LLMs enhance DHS's threat intelligence by aggregating and interpreting data from diverse sources, including external cyber threat feeds, previous incidents, and ongoing system monitoring. This constant stream of information allows AI to predict emerging threats and recognize sophisticated cyberattack tactics. The integration of predictive analytics into threat intelligence empowers DHS to make more informed, data-driven decisions, enabling a faster response to evolving cyber threats ([A Comprehensive Overview of Large Language Models \(LLMs\) for Cyber Defences: Opportunities and Directions](#)).

Enhancing Cybersecurity Readiness Through AI Insights

AI-driven systems continuously analyze both historical and real-time data, improving DHS's readiness against cyberattacks. By identifying vulnerabilities and offering mitigation strategies, AI tools can significantly reduce the time needed to address emerging threats. LLMs can also suggest proactive defense strategies, ensuring that DHS remains ahead of potential cyber adversaries. The use of AI insights in decision-making strengthens the agency's ability to anticipate risks and fortify its cybersecurity infrastructure ([Leveraging AI to Enhance the Nation's Cybersecurity](#))

Future-Proof Security with AI-Powered Defenses

As cyber threats continue to evolve, it is essential for organizations, including the Department of Homeland Security (DHS), to adopt adaptive cybersecurity strategies that can continually evolve to counter new threats. Artificial intelligence (AI), specifically machine learning and Large Language Models (LLMs), can significantly enhance these strategies, enabling DHS to not only defend against existing threats but also to anticipate and mitigate emerging risks. This section outlines how AI can help future-proof DHS's security infrastructure by enabling adaptive strategies, fostering continuous learning, and strengthening long-term national security.

Adaptive Cybersecurity Strategies Using AI.

AI systems, particularly machine learning models, can adapt to new and evolving threats by continuously analyzing incoming data and recognizing new attack patterns. By incorporating AI into DHS's cybersecurity strategy, the agency can ensure that its defenses are always aligned with the latest threat landscape. As cyber adversaries adopt more sophisticated tactics, AI-powered systems can dynamically adjust their defense mechanisms to stay ahead of evolving attack vectors, providing DHS with a robust and flexible security posture ([Leveraging AI to Enhance the Nation's Cybersecurity](#)).

Continuous Learning Models to Counter Evolving Threats

The ability of AI models to learn from vast amounts of data and improve over time is crucial for combating increasingly sophisticated cyber threats. These models can be continuously trained on historical data, threat intelligence feeds, and real-time network activity, allowing them to better identify and mitigate future threats. By fostering continuous learning, AI can help DHS stay resilient against novel attacks and emerging tactics, reducing the risk of vulnerabilities being exploited ([AI in cybersecurity](#))

Strengthening DHS's Long-Term National Security Posture

AI-powered defenses provide DHS with a long-term solution to strengthening the nation's cybersecurity infrastructure. As new technologies and attack methodologies emerge, AI models can quickly adapt and mitigate risks, ensuring that DHS's defenses remain relevant and effective in the face of ever-changing threats. By investing in AI-driven security solutions, DHS can safeguard critical infrastructure, protect sensitive data, and maintain a proactive stance against both current and future cyber threats ([Leveraging AI to Enhance the Nation's Cybersecurity](#)).

Conclusion

As cybersecurity challenges grow more complex and the volume of data continues to increase, the integration of Artificial Intelligence (AI), particularly Large Language Models (LLMs), becomes an essential tool for enhancing the Department of Homeland Security's (DHS) ability to protect the nation's critical infrastructure and sensitive information. AI has the potential to significantly improve efficiency in cybersecurity operations, alleviating staff workload by automating time-consuming tasks such as threat detection, log analysis, and incident response.

The adoption of AI-driven solutions can enhance DHS's capabilities in predictive intelligence, proactive threat mitigation, and continuous learning. By analyzing historical data and real-time activities, AI models empower DHS to stay ahead of evolving cyber threats, reducing response times and improving the overall security posture. Furthermore, AI can assist in regulatory compliance, ensuring adherence to federal cybersecurity policies and simplifying audit processes.

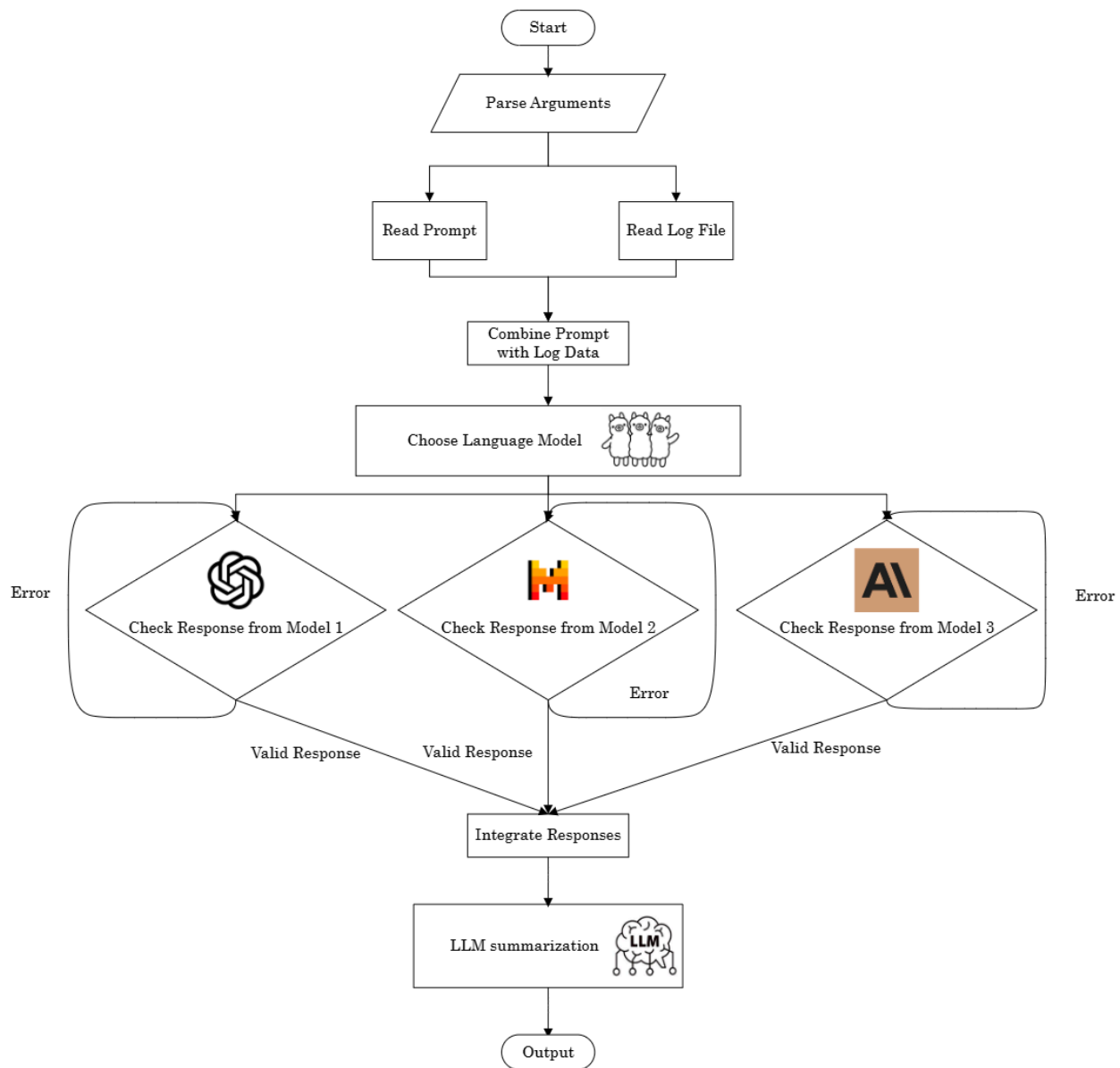
The use of AI within DHS operations is not only crucial for addressing current security concerns but also for future-proofing the agency's cybersecurity defenses. AI's adaptive nature ensures that DHS will remain resilient against the continuously evolving cyber threat landscape, offering a dynamic approach to national security.

To remain competitive and secure in an increasingly complex cyber environment, DHS must continue to embrace and integrate AI-driven technologies into its security strategies. The successful implementation of AI will reduce the workload of cybersecurity personnel, enhance operational efficiency, and fortify the agency's ability to counteract both current and emerging cyber threats.

In summary, AI-driven cybersecurity solutions are not just a luxury, but a necessity for modernizing DHS operations. The integration of AI into threat detection, log analysis, and incident response processes will significantly reduce human error, improve accuracy, and ensure that DHS is better equipped to protect national security in the face of an ever-growing threat landscape.

Appendix

Here, we provide **an example of how we can use LLM to facilitate log analysis**. We created a Python script, which functions as an automated log analysis tool that leverages multiple LLMs to process log files, extract key insights, and generate a structured summary. The overall workflow is shown below:



First, the script reads a given log file and combines them with a predefined prompt from a YAML configuration file. Then, it sends this data to different LLMs via HTTP requests using the Ollama API. The script supports multiple local models, such as Llama3, Mistral, and Gemma, and processes them in parallel, which allows efficient execution. Each model's response is logged and stored in a model-specific output file to ensure traceability. The example logs and the responses from different LLMs are shown below:

```

1 Dec 30 00:01:02 Jin CRON[999999]: pam_unix(cron:session): session closed for user project
2 Dec 30 00:01:36 Jin sshd-session[999991]: Connection closed by 101.43.42.253 port 40827
3 Dec 30 00:05:01 Jin sshd-session[999992]: reprocess config line 33: Deprecated option RSAAuthentication
4 Dec 30 00:05:01 Jin sshd-session[999992]: reprocess config line 40: Deprecated option RhostsRSAAuthentication
5 Dec 30 00:05:01 Jin sshd-session[999992]: Failed password for project from 10.0.42.195 port 57106 ssh2
6 Dec 30 00:05:01 Jin sshd-session[999992]: Connection closed by authenticating user project 10.0.42.195 port 57106 [preauth]
7 Dec 30 00:05:01 Jin sshd-session[999993]: reprocess config line 33: Deprecated option RSAAuthentication
8 Dec 30 00:05:01 Jin sshd-session[999993]: reprocess config line 40: Deprecated option RhostsRSAAuthentication
9 Dec 30 00:05:01 Jin sshd-session[999993]: Accepted password for project from 147.12.82.196 port 57108 ssh2
10 Dec 30 00:06:36 Jin sshd-session[999994]: Connection closed by 192.168.42.20 port 38275
11 Dec 30 00:10:01 Jin sshd-session[999995]: reprocess config line 33: Deprecated option RSAAuthentication
12 Dec 30 00:10:01 Jin sshd-session[999995]: reprocess config line 40: Deprecated option RhostsRSAAuthentication
13 Dec 30 00:10:03 Jin sshd-session[999995]: Accepted publickey for admin from 10.1.42.12 port 6234 ssh2: RSA SHA256:ABCD1234XYZ
14 Dec 30 00:10:07 Jin sshd-session[999996]: Failed password for root from 221.10.11.111 port 52222 ssh2
15 Dec 30 00:10:08 Jin sshd-session[999996]: Connection closed by authenticating user root 221.10.11.111 port 52222 [preauth]
16 Dec 30 00:15:15 Jin sshd-session[100000]: Accepted publickey for dev from 8.8.8.8 port 50022 ssh2: RSA SHA256:ZZZZZZZZZZZZ

```

```
(base) PS F:\DASSH 2025\llm-log-analyzer-main> python log-analyzer.py -f test-auth.log -c prompt.yaml -models llama3.2 mistral gemma:7b --summarize
```

```

Starting the LLM log analyzer with multiple models
-> Sending POST request to Ollama using model: llama3.2
-> Sending POST request to Ollama using model: mistral
-> Sending POST request to Ollama using model: gemma:7b

```

```
=====
LLM RESPONSE FROM LLAMA3.2:
```

```
=====
Suspicious behavior identified:
```

1. Multiple occurrences of deprecated options 'RSAAuthentication' and 'RhostsRSAAuthentication' on the same config line, indicating a potential misuse or misunderstanding of SSH configuration options.
2. A successful authentication attempt with an unknown username 'root' from an external IP address '221.10.11.111', which may indicate a brute-force or unauthorized access attempt.
3. A connection closure by a user who has already authenticated successfully ('authenticating user root'), suggesting potential false shutdowns or abuse of the SSH service.

```
Anomalies and potential threats:
```

1. Unusual login attempts from unknown IP addresses (147.12.82.196, 221.10.11.111).
2. Repeated authentication failures for an unauthorized username.
3. Connection closures by authenticated users.
4. Successful public key authentication with an unusual hash value ('ZZZZZZZZZZZZ') and an external IP address ('8.8.8.8').

```
=====
LLM RESPONSE FROM MISTRAL:
```

1. The first two lines indicate that the SSH daemon on the system is attempting to reprocess deprecated options in its configuration file (RSAAuthentication and RhostsRSAAuthentication). While this by itself is not necessarily malicious, it could be an indication of an older or misconfigured SSH server.
2. The third line shows a successful login using the password for the user 'project' from IP address 147.12.82.196. This is normal behavior if the user has been granted the necessary permissions to log in.
3. The sixth and seventh lines show a failed login attempt for the user 'root' from IP address 221.10.11.111, followed by the connection being closed. This could potentially be a brute-force attack or an attempted privilege escalation.
4. The ninth line shows another successful login using a public key for the user 'dev' from IP address 8.8.8.8. Google's public DNS servers (8.8.8.8 and 8.8.4.4) are often used by legitimate traffic, but without more context or knowledge about the specific environment, it is difficult to say if this is suspicious or not.
5. The tenth line indicates another successful login using a public key for the user 'admin' from IP address 10.1.42.12. This is normal behavior if the user has been granted the necessary permissions to log in.

To better improve the workflow, our tool also includes an optional summarization feature, where an additional LLM aggregates responses from multiple models into a concise summary including key insights and discrepancies between outputs. This process is particularly useful for handling large log files, as it automates what would otherwise be a tedious manual review. A sample summarization is provided below:

```

Generating summary of all model responses...
-> Generating summary using llama3...

=====
SUMMARY OF ALL RESPONSES (GENERATED BY LLAMA3):
=====
Here is the comprehensive summary of the log file analysis from multiple LLM models:

Key Insights:

1. The SSH daemon on the system is attempting to reprocess deprecated options in its configuration file (RSAAuthentication and RhostsRSAAuthentication), indicating potential misuse or misunderstanding of SSH configuration options.
2. There are suspicious login attempts and connection closures that may indicate brute-force or unauthorized access attempts, as well as potential false shutdowns or abuse of the SSH service.
3. Unusual public key authentication with an unusual hash value ('ZZZZZZZZZZ') and an external IP address ('8.8.8.8') was detected.

Important Patterns:

1. Repeated occurrences of deprecated options 'RSAAuthentication' and 'RhostsRSAAuthentication' on the same config line, indicating potential misuse or misunderstanding of SSH configuration options.
2. Successful authentication attempts with unknown usernames from external IP addresses, which may indicate brute-force or unauthorized access attempts.
3. Connection closures by authenticated users, suggesting potential false shutdowns or abuse of the SSH service.

Significant Discrepancies:

1. LLAMA3.2 and GEMMA:7B both identify suspicious password authentication for the 'project' user from an external IP address (147.12.82.196), while MISTRAL does not mention this finding.
2. LLAMA3.2 and GEMMA:7B both identify a failed root login attempt, while MISTRAL only mentions a successful login using the password for the user 'project' from the same IP address.

Overall, the log file analysis suggests potential security threats and misuse of SSH configuration options, as well as unusual public key authentication attempts.
=====

```

In this example, **the LLMs serve two primary roles**: first, as **analysts**, they process raw log data to detect patterns, anomalies, and insights that may indicate system performance issues, security threats, or operational inefficiencies. Second, as **summarizers**, they refine and condense the extracted insights into a high-level overview to remove redundant information and ensure clarity. The script also implements error handling and logging to make it robust for real-world deployment. If any model fails to return a valid response, errors are logged, and execution continues without disruption.

By integrating LLMs into log analysis workflows, our example shows the potential of LLMs and AI to provide a scalable and intelligent method to analyze logs, particularly in fields like cybersecurity, system monitoring, and AI-assisted IT operations. The use of LLMs effectively turns unstructured log data into meaningful reports and allows engineers and decision-makers to focus on actionable insights rather than spending time parsing raw logs.

```

ALERT: BLOCK detected from 192.168.1.50
Password:
sudo: iptables: command not found
ALERT: FAILED_LOGIN detected from 192.168.1.100
Alert: Unusual logins from 192.168.1.100
ALERT: ALERT detected from 192.168.1.150
System 192.168.1.150 quarantined

```

Plain Text



Language

Run

Stop

Run Settings...

logs.json

process_logs.py

```
1  [
2    {
3      "timestamp": "2025-03-01 12:35:22",
4      "source": "Firewall",
5      "event_type": "BLOCK",
6      "ip_address": "192.168.1.50",
7      "message": "Malicious Traffic Detected"
8    },
9    {
10     "timestamp": "2025-03-01 12:36:10",
11     "source": "Authentication",
12     "event_type": "FAILED_LOGIN",
13     "ip_address": "192.168.1.100",
14     "message": "Multiple failed login attempts detected"
15   },
16   {
17     "timestamp": "2025-03-01 12:37:05",
18     "source": "System",
19     "event_type": "ERROR",
20     "ip_address": "N/A",
21     "message": "Kernel Panic - System halted unexpectedly"
22   },
23   {
24     "timestamp": "2025-03-01 12:38:30",
25     "source": "Firewall",
26     "event_type": "ALLOW",
27     "ip_address": "203.0.113.45",
28     "message": "Normal network traffic"
29   },
30   {
31     "timestamp": "2025-03-01 12:39:15",
32     "source": "IDS",
33     "event_type": "ALERT",
34     "ip_address": "192.168.1.150",
35     "message": "Suspicious port scanning activity detected"
36   }
37 ]
38
```

